ASTANA IT UNIVERSITY

«Approved»
by the Dean
________ Syzdykova Z.A.
«___» of _________ 2024

Syllabus
Academic Year 2024-2025

1. General Information	
Course Code	
Course title	**Linear Algebra**
Degree cycle (level)/ major	Bachelor, Group of IT specialties, Software Engineering, Information technology
Year, semester	1, 1
Number of credits	5
Language of Delivery	English
Prerequisites	School mathematics course
Postrequisites	Calculus 2, Probability and Statistics, Algorithms and Data structures, Advanced programming
Lecturer(s)	Abdygali Jandigulov, the Candidate of Physical-Mathematical Sciences, Associate Professor, a.jandigulov@astanait.edu.kz, Astana IT University, Expo, C1 block, 2nd floor, office C1.1.335).
2. Goals, Objectives and Learning Outcomes of the Course	
1. Course Description	The course covers systems of linear equations, matrices, inverse of a matrix, determinant, vectors in two-, three- and n-dimensions, vector spaces and subspaces, eigenvectors and eigenspaces, and applications of linear algebra
2. Course Goal(s)	To provide students with the knowledge of fundamentals of linear algebra and the theory of matrices. On completion of this course the student will master the basic concepts and acquires skills in solving problems in linear algebra
3. Course Objectives	Comprehend vector spaces and subspaces. Understand fundamental properties of matrices including determinants, inverse matrices, matrix factorizations, eigenvalues, and linear transformations. Solve linear systems of equations
4. Skills & Competences	ability to write mathematical statements and problem solutions using mathematical symbols; understanding of key mathematical concepts and the application of appropriate tools to real problems; Writing logical progressions of precise mathematical statements to justify and communicate your reasoning
5. Course Learning Outcomes	By the end of this course students will be able to: Use Gauss-Jordan elimination to solve systems of linear equations and to compute the inverse of an invertible matrix. Use the basic concepts of vector and matrix algebra, including linear dependence / independence, basis and dimension of a subspace, rank, and nullity, for analysis of matrices and systems of linear equations. Evaluate determinants and use them to discriminate between singular and nonsingular matrices.

	Use the characteristic polynomial to compute the eigenvalues and eigenvectors of a square matrix and use them to diagonalize matrices when this is possible. Combine methods of matrix algebra to compose the change-of-basis matrix with respect to two bases of a vector space
6. Methods of Assessment	The expected learning outcomes for the course will be assessed through graded activities and ungraded activities. The graded activities include exams, homework assignments, and quizzes. The ungraded activities will include a monitoring of attendance and class participation. A variety of ungraded assessment techniques may be employed, that include problems to be completed during class, direct questioning of students, answering student's questions in class, and discussions during office hours
7. Type of teaching	**Lectures** serve to introduce new concepts and provide theoretical and methodological foundations. **Practice sessions (seminars)** are active sessions to develop student's confidence through new examples and discussions on the problems of higher education and didactics. **Instructor-supervised independent study (ISIS)** focuses on the review of reviewing research papers, theories, and practices. It is designed to explores in greater depth of the course material. **Student's independent study (SIS):** Self-study time including the time required to prepare for and complete all course assignments. **Teaching methods:** Formal lecture, semi-formal lecture, multimedia lecture, active learning, interrogative teaching methods, discussion, colloquium
8. Reading List	Assigned reading materials and presentations should be read prior to class. Class lectures and discussions will proceed with supplemental and advanced topics, which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and / or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu.kz). Basic literature: **1.** Linear Algebra by Jonathan Gleason, AMS Open Math Notes: Works in Progress; Last Revised: 2018-06-04. https://www.ams.org/open-math-notes/omn-view-listing?listingId=110772 **2.** Linear Algebra and Matrix Analysis for Statistics. by Sudipto Banerjee, Anindya Roy, Released June 2014. Publisher(s): Chapman and Hall/CRC. ISBN: 9781482248241. https://www.oreilly.com/library/view/linear-algebra-and/9781482248241/ **3.** *Linear Algebra with Applications* by W. Keith Nicholson. An Open Textbook. 2023. https://lyryx.com/linear-algebra-applications/ **4.** MATH 233 - Linear Algebra I Lecture Notes Cesar O. Aguilar. https://pdfprof.com/PDF_DocsV2/Documents/144673/5/5 Supplementary literature: 1. Линейная алгебра в задачах и упражнениях: Учебное пособие. — 3 е изд., испр. — СПб.: Издательство «Лань», 2016 — 592 с. — (Учебники для вузов. Специальная литература). https://elar.urfu.ru/bitstream/10995/78551/1/978-5-7996-2776-8_2019.pdf

	2. Сборник задач по линейной алгебре: Учебное пособие. 13-е изд., стер. — СПб.: Издательство «Лань», 2010. — 480 с. — (Учебники для вузов. Специальная литература) http://lib.ysu.am/open_books/418693.pdf
9. Resources	Online journals, article, papers, books, and internet resources as well as online emulators and online software for simulation
10. Course **policy**	**Course and University policies include:** **Attendance:** Attendance is not allocated any grading points in the marking scheme but is compulsory to pass the course. Normally students are required to achieve course attendance of minimum 70% to get admitted to the examination rubric. In case a student misses 30% or more class sessions without a valid excuse the instructor has the right to mark him as "not graded". In such case a student is not admitted to the exam and automatically fails the course. It should be NOTED that in cases when a student is excused for 30% of the scheduled class sessions or more, he or she must study material provided under the course on their own. Course instructor might provide additional opportunities to submit missed graded pieces of work during office hours or conduct alternative assessment exercises using method of his or her choosing. **Preparation for Class**: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties: - Offers a different and unique, but relevant, perspective; - Contributes to moving the discussion and analysis forward; - Builds on other comments. **Class work**: The duration of each lecture and practical lesson is 50 minutes for offline class, and 40 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student's misconduct, the student can be dispensed from the classes. **Being late on class:** When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, the being late to the class is not welcome and can have restriction activities by the course instructor. **Attestation I and II:** Students, who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course.

	Homework / Assignments: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. Assigned: HW will be assigned every week. HW assignments include problem sets and propositions to prove. Reading: It is the students' responsibility to read carefully and thoroughly the appropriate section(s) from the textbooks and review their class notes after each class. **Late submissions:** Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All gradings are based using a percentage grading scale. *In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances by the instructor.* **Final exam** will be held on the Examination session (according to the Academic Calendar). Form of the Final exam is written exam. **Laptops and mobile devices** can only be used for classroom purposes when directed by the course instructor. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the instructor. **Online lessons** can be used in case if there won't be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround. **Cheating and plagiarism** are defined in the Academic conduct policies of the university and include: 1. Submitting work that is not your own papers, assignments, or exams; 2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation; 3. Submitting or using falsified data; 4. Submitting the same work for credit in two courses without prior consent of both instructors. Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university. **Academic Conduct Policies of the university:** The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu.kz). **Contacting the Course instructor**: The easiest and most reliable way to get in touch with the course instructor is by email. Students must feel free to send email if you have a question related to the course. Instructor responds as soon as they can but not always instantaneously. Besides that, students are also welcomed to arrange a one-to-one meeting with the instructor by their office during office hours to discuss the class using both offline and online

3. Course Content

3.1 Lecture, Practical/Seminar/Laboratory Session Plans

Week No	Course Topic	Lectures (Hours/Week)	Practice sessions (H/W)	TSIS (H/W)	SIS (H/W)
1	**Linear Equations in Linear Algebra** Systems of Linear Equations; Row Reduction and Echelon Forms; Vector Equations; The Matrix Equation; Solution Sets of Linear Systems	3	2	1	9
2	**Linear Independence**; Introduction to Linear Transformations; The Matrix of a Linear Transformation	3	2	1	9
3	**Matrix Algebra** Matrix Operations; The Inverse of a Matrix; Characterizations of Invertible Matrices; Matrix Factorizations	3	2	1	9
4	**Determinants.** Introduction to Determinants; Properties of Determinants; Cramer's Rule, Volume	3	2	1	9
5	**Vector Spaces**. Vector Spaces and Subspaces; Subspaces of R^n; Null Space, Column Space, Row space	3	2	1	9
6	Linearly Independent Sets; Bases; Coordinate Systems; The Dimension of a Vector Space; Rank; Change of Basis	3	2	1	9
7	**Eigenvalues and Eigenvectors.** The Characteristic Equation; Diagonalization	3	2	1	9
8	Eigenvectors and Linear Transformations; Complex Eigenvalues	3	2	1	9
9	**Orthogonality.** Inner product; length, and orthogonality; Orthogonal sets; Orthogonal projections	3	2	1	9
10	Review of the course	3	2	1	9
	Total hours:	**30**	**20**	**10**	**90**

3.2 List of Assignments for Student Independent Study

№	Assignments (topics) for independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	2	3	4	5
1	**Chapter 1 Linear Equations in Linear Algebra** 1.1 Systems of Linear Equations; 1.2 Row Reduction and Echelon Forms; 1.3 Vector Equations; 1.4 The Matrix Equation; 1.5 Solution Sets of Linear Systems	9	[1-4]	Handwritten reports of problem sets
2	1.7 **Linear Independence**; 1.8 Introduction to Linear Transformations; 1.9 The Matrix of a Linear Transformation	9	[1-4]	Handwritten reports of problem sets
3	**Chapter 2 Matrix Algebra** 2.1 Matrix Operations; 2.2 The Inverse of a Matrix; 2.3 Characterizations of Invertible Matrices; 2.4 Partitioned Matrices; 2.5 Matrix Factorizations	9	[1-4]	Handwritten reports of problem sets
4	**Chapter 3 Determinants** 3.1 Introduction to Determinants; 3.2 Properties of Determinants; 3.3 Cramer's Rule, Volume, and Linear Transformations	9	[1-4]	Handwritten reports of problem sets
5	**Chapter 4 Vector Spaces** 4.1 Vector Spaces and Subspaces; 2.8 Subspaces of R^n; 4.2 Null Spaces, Column Spaces, Row space **Midterm exam**	9	[1-4]	Handwritten reports of problem sets, and preparation to quiz and exams
6	4.3 Linearly Independent Sets; Bases. 4.4 Coordinate Systems. 4.5 The Dimension of a Vector Space; 4.6 Rank; 4.7 Change of Basis	9	[1-4]	Handwritten reports of problem sets
7	**Chapter 5 Eigenvalues and Eigenvectors** 5.1 Eigenvectors and Eigenvalues; 5.2 The Characteristic Equation; 5.3 Diagonalization	9	[1-4]	Handwritten reports of problem sets
8	5.4 Eigenvectors and Linear Transformations; 5.5 Complex Eigenvalues.	9	[1-4]	Handwritten reports of problem sets
9	**Chapter 6. Orthogonality.** 6.1 Inner product; length, and orthogonality; 6.2 Orthogonal sets; 6.3 Orthogonal projections.	9	[1-4]	Handwritten reports of problem sets
10	Review of the course **Endterm exam**	9	[1-4]	Handwritten reports of problem sets, and preparation to quiz and exams
	Total:	**90**		

4. Student Performance Evaluation System for the Course

assessment and forms of examination Course	Period	Assessment type	Number of points	Exam Form	Schedule (Week #)
	1st attestation	Problem Sets	40	Submission of written reports	Weekly
		Assignment 1	10	Online submission in to the Moodle	5th week
		Mid-term Exam	50	Written	5th week
		1st attestation total	**100**		
	2nd attestation	Problem Sets	40	Submission of written reports	Weekly
		Assignment 2	10	Online submission in to the Moodle	10th week
		End-term Exam	50	Written	10th week
		2nd attestation total	**100**		
Final exam	A written exam				100
Total	**0,3*1stAtt+0,3*2ndAtt+0,4*Final**				**100**

Period	**Assignments**	**Number of points**	**Total**
1st attestation	Problem set-1 Problem set-2 Assignment-1 **Mid Term Exam**	20 20 10 50	100
2nd attestation	Problem set-3 Problem set-4 Assignment-2 **End Term Exam**	20 20 10 50	100
Final exam	Written Exam		100
Total	**0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final**		**100**

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	**Numerical equivalent**	**Percentage**	**Grade according to the traditional system**
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	Satisfactory
C-	1,67	60-64	
D+	1,33	55-59	
D	1,0	50-54	
FX	0	25-49	Fail
F	0	0-24	

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment
80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement
60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught
25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught
0-24	No significant assessable material, absent or assessment missing a must pass component

5. Methodological Guidelines

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **ISIS (Instructor Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- **Final assessment** is a combination of both individual (team) project (report) and oral presentation (slides) to evaluate the students' academic performance and professional skills. At the completion of this course each student must submit an online project version conforming to the project outline, as well as to prepare and present a slide presentation that follows the presentation outline. Project iterations would be required to submit as well.

Students should submit the written Report of Final project and slide-Presentation 3 days before the Final exam day.

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Abdygali Jandigulov	Associated professor	26.08.24	
Ainur Basheyeva	Associated professor	26.08.24	

Director **Sergaziyev M.Zh.**